

Zander Carpio

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Mathematics undergraduate working on quantitative and analytical problems: epidemic dynamics, geospatial disease surveillance, official statistics, and computational number theory. Seeking analytical, data and research roles.

Education

BS Mathematics Aug 2023 to 2027 (expected)

University of the Philippines Baguio

DOST-SEI Undergraduate Scholar, a competitive national merit scholarship from the Philippine Department of Science and Technology. Undergraduate thesis in number theory.

Relevant coursework: Real Analysis, Complex Analysis, Analytic Number Theory, Operations Research, Physics, Accounting.

Secondary Education 2017 to 2023

Aurora National Science High School

Experience

Intern, Statistical Operations and Coordination Division Jun to Jul 2026

Philippine Statistics Authority, Baguio City

- Statistical operations, data visualisation and coordination work inside the national statistics agency responsible for official statistics, the census and civil registration.
- Worked with survey and administrative data in its delivered state, including inconsistent labelling and failed joins.

Peer Mentor, Learning Resource Center Sep 2025 to present

University of the Philippines Baguio, Contract

Mentoring undergraduates through mathematics coursework.

Mathematics Tutor Jul 2021 to Jul 2024

Freelance, Hybrid

Three years tutoring mathematics at secondary and lower undergraduate level.

Social Media Manager Jul 2022 to present

Freelance, Remote

Four years running social media for small businesses alongside a full-time degree. Campaign management, scheduling and reporting in GoHighLevel and SocialPilot, with graphic design and content produced in house. Client names withheld.

Quantitative Projects

Dengue surveillance analysis, Baguio City R, QGIS

Reconstructed barangay-level surveillance across 127 barangays and 23 reporting weeks, joined on PSGC codes and mapped as a choropleth. Showed the burden is concentrated rather than citywide, the worst tenth of barangays carrying 32% of cases, and confirmed this was not an artefact of barangay size by re-deriving it from population-normalised incidence (32.4% against 31.7%). That check exposed a denominator failure in the commercial core, where residential population understates the at-risk daytime population enough to yield nominal rates above 4,000 per 1,000.

Compartmental epidemic models R, deSolve, Shiny

SIR and SEIR systems with mass-action incidence, integrated with **deSolve** and published as interactive Shiny applications in which the parameters can be varied and the epidemic curve re-solved.

Cholera transmission with an environmental reservoir R, MATLAB

SEIRB model coupling direct transmission to a bacterial reservoir, built as a 797-line Shiny application with fourteen tunable parameters and cross-implemented in MATLAB.

Kaprekar triples, undergraduate thesis PARI/GP, LaTeX

Extending Iannucci and Foster (2005). Reduced an infinite search to a finite one by proving every triple is a root of a congruence, bounding the count at four per power of ten. Worked to a rule that no result counts until two independently written programs agree, which exposed a counting error in earlier code.

Technical Skills

Languages	R, Python, MATLAB, C, PARI/GP
Libraries	deSolve, Shiny, ggplot2, dplyr, tidyr, NumPy
Tools	QGIS, LaTeX, Git, Excel, GoHighLevel, SocialPilot